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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Statistical Foundation for Big Data Analysis (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 0 ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

# Week 9 : Assignment 9

Assignment not submitted **Due date: 2026-03-25, 23:59 IST.**

**Instructions:**

- Each question in this section has **exactly one correct option**.
- Candidate must select the correct option.
- **Each question carries 2 marks**.
- **There is no negative marking** for incorrect answers.
- Marks will be awarded **only if the correct option is clearly selected**.
- All questions are **compulsory**.

1) ***2 points***

Let $A$ be an $n \times n$ matrix with real entries. Which of the following statements are correct?

(a) If $\mathbf{v}$ is an eigenvector of $A$ corresponding to eigenvalue $\lambda$, then $c\mathbf{v}$ is also an eigenvector for any scalar $c \neq 0$.

(b) The difference of two eigenvectors corresponding to the different eigenvalues is always an eigenvector.

(c) The sum of two eigenvectors corresponding to different eigenvalues is always an eigenvector.

(d) Every eigenvector must be of unit length.

- a.
- b.
- c.
- d.

**Week 9 ()**

- Lecture 41 : Introduction to Principal Component Analysis (unit?unit=101&lesson=103)
- Lecture 42 : Eigenvalues and Eigenvectors (unit?unit=101&lesson=104)
- Lecture 43 : Mathematical Formulation of Principal Component Analysis (unit?unit=101&lesson=105)
- Lecture 44 : Mathematical Foundation of Principal Component Analysis (unit?unit=101&lesson=106)
- Lecture 45 : Population Principal Component Analysis for Normal Distribution (unit?unit=101&lesson=107)
- **Quiz: Week 9 : Assignment 9 (assessment?name=102)**
- Feedback Form (unit?unit=101&lesson=108)

2) ***2 points***

Let $X = (X_1, X_2, X_3)^T$ be a random vector with mean zero and population covariance matrix after centering is given by:

$$\Sigma = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Which of the following is the proportion of total population variance explained by the first principal component?

(a) $\frac{4}{6}$

(b) $\frac{4}{7}$

(c) $\frac{2}{7}$

(d) $\frac{1}{4}$

- a.
- b.
- c.
- d.

3) ***2 points***

Let $A$ be an $n \times n$ matrix with real entries. Which of the following statements is always true?

(a) The sum of the eigenvalues of $A$ equals $\det(A)$ (the determinant of the matrix $A$).

(b) The product of all the eigenvalues of $A$ equals the sum of all the diagonal entries of the matrix $A$.

(c) The sum of all the eigenvalues of $A$ equals the sum of all the diagonal entries of the matrix $A$.

(d) All eigenvalues of $A$ are positive.

- a.
- b.
- c.
- d.

Lecture Material (unit?unit=101&lesson=109)

LIVE SESSION ()

**Instructions:**

- Each question in this section may have **one or more correct options**.
- Candidates must **select all and only the correct options** for each question.
- **Each question carries 2 marks**, which will be awarded **only if the exact set of correct answers is marked**.
- **Partial marking is not allowed**. Any missing correct option or selection of an incorrect option will result in **zero marks** for that question.
- **There is no negative marking** for incorrect answers.
- Marks will be awarded **only when the response exactly matches the correct answer set**.
- All questions are **compulsory**.

4) ***2 points***

Which statements correctly describe Principal Component Analysis (PCA)?

(a) PCA reduces dimensionality while preserving directions of maximum variance.
(b) PCA removes variance from the data.
(c) PCA can improve computational efficiency and reduce overfitting.
(d) PCA enables visualization in lower dimensions.

- [ ] a.
- [ ] b.
- [ ] c.
- [ ] d.

5) ***2 points***

Which statements correctly describe the covariance matrix in Principal Component Analysis?

(a) Diagonal entries represent variance of individual features.
(b) Off-diagonal entries represent covariance between features.
(c) Covariance depends on the mean being subtracted from the data.
(d) Covariance matrix is always identity for centered data.

- [ ] a.
- [ ] b.
- [ ] c.
- [ ] d.

6) ***2 points***

Which of the following statements about the characteristic polynomial $p_A(t)$ $\det(A - tI)$ of an $n \times n$ matrix $A$ with real entries are true?

(a) Its roots give the eigenvalues of $A$.
(b) The matrix $A - \lambda I$ is singular if $\lambda$ is an eigenvalue of $A$.
(c) All eigenvalues of $A$ must be real.
(d) The degree of $p_A(t)$ is always $n$.

- ☐ a.
- ☐ b.
- ☐ c.
- ☐ d.

7) **2 points**

Which of the following statements about diagonalization and positive definite matrices are correct?

(a) A matrix with $n$ linearly independent eigenvectors is diagonalizable.
(b) A real symmetric matrix can always be diagonalized by an orthogonal matrix.
(c) Every positive definite matrix has all positive eigenvalues.
(d) Only matrices with distinct eigenvalues are diagonalizable.

- ☐ a.
- ☐ b.
- ☐ c.
- ☐ d.

8) **2 points**

Which statements about projected distances along the first principal component (PC1) are correct?

(a) They capture the spread of the data along the direction of maximum variance.
(b) Their variance is always equal to the total variance of the dataset.
(c) They are the principal component that has maximum variance.
(d) The squared projected distances are related to the corresponding eigenvalue of PC1.

- ☐ a.
- ☐ b.
- ☐ c.
- ☐ d.

9) **2 points**

Which statements correctly describe the challenges of high-dimensional data?

(a) High-dimensional data can lead to overfitting.
(b) High-dimensional data always improves model generalization.
(c) High-dimensional data can be difficult to visualize.
(d) High-dimensional data increases sparsity and reduces distance informativeness.

- ☐ a.
- ☐ b.
- ☐ c.
- ☐ d.

**Instructions:**

- This section consists of **numerical answer type** questions.
- Candidates must **write the final numerical answer only**.
- **Each question carries 2 marks**, which will be awarded **only if the answer is correct**.
- **No partial marks** will be awarded for intermediate steps or partially correct answers.
- Answers must be **rounded off as specified** in the question; answers not rounded as required will be treated as incorrect.
- **There is no negative marking** for incorrect answers.
- All questions are **compulsory**.

10)

Let $A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$ be a $3 \times 3$ matrix with eigen values $\lambda_1, \lambda_2$ and $\lambda_3$ such that $\lambda_1 > \lambda_2 > \lambda_3$. Then, find the value of $\lambda_1$ (correctly upto 3 decimal places).

***2 points***

You may submit any number of times before the due date. The final submission will be considered for grading.

**Submit Answers**